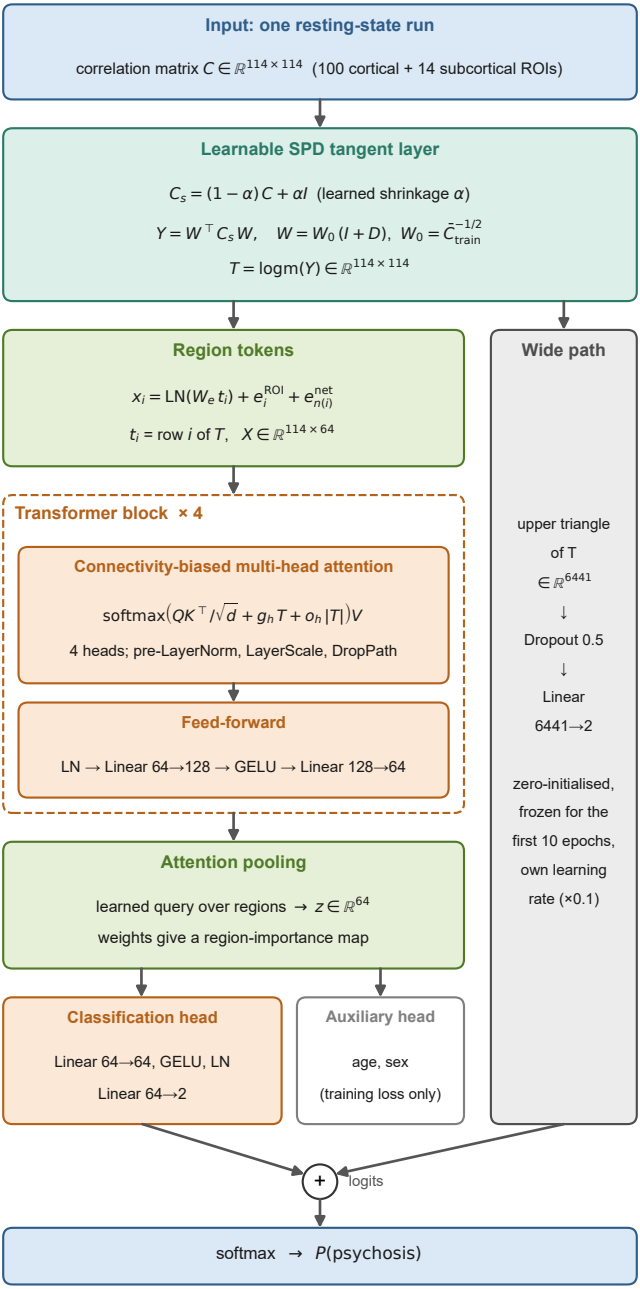

Input: one resting-state run
correlation matrix $C \in \mathbb{R}^{114\times114}$ (100 cortical + 14 subcortical ROIs)
Learnable SPD tangent layer
$C_s = (1-\alpha)\,C + \alpha I$ (learned shrinkage α)
$Y = W^\top C_s W, \quad W = W_0\,(I + D), \; W_0 = \bar{C}_{\text{train}}^{-1/2}$
$T = \operatorname{logm}(Y) \in \mathbb{R}^{114\times114}$
Region tokens
$x_i = \mathrm{LN}(W_e\, t_i) + e_i^{\text{ROI}} + e_{n(i)}^{\text{net}}$
t_i = row i of T, $X \in \mathbb{R}^{114\times64}$
Transformer block × 4
Connectivity-biased multi-head attention
$\mathrm{softmax}\big(QK^\top/\sqrt{d} + g_h T + o_h |T|\big)V$
4 heads; pre-LayerNorm, LayerScale, DropPath
Feed-forward
LN → Linear 64→128 → GELU → Linear 128→64
Attention pooling
learned query over regions → $z \in \mathbb{R}^{64}$
weights give a region-importance map
Classification head
Linear 64→64, GELU, LN
Linear 64→2
Auxiliary head
age, sex
(training loss only)
Wide path
upper triangle of T $\in \mathbb{R}^{6441}$
↓
Dropout 0.5
↓
Linear 6441→2
zero-initialised, frozen for the first 10 epochs, own learning rate (×0.1)
+
logits
softmax → P(psychosis)